

# Engineering Remediation Playbook

## Overview of Affected Assets

This playbook addresses two primary concerns:

1. **AWS Emulator Unreachability:** The AWS Emulator is unreachable on port 4566, causing issues with S3 bucket interactions.
2. **Public S3 Bucket Permissions:** Public S3 buckets lack adequate permissions, exposing sensitive data to unauthorized parties.

## Step-by-Step Remediation Commands

### 1. Enable AES-256 Server-Side Encryption

**Description:** Encrypt unencrypted S3 buckets using FIPS-compliant algorithms for added security.

**Command:**

```
aws s3api put-bucket-encryption --bucket <BUCKET_NAME> --server-side-encryption-Type AES_256
```

Replace `<BUCKET_NAME>` with the name of your target bucket.

### 2. Enable Public Access Blocks

**Description:** Restrict public access to S3 buckets using ACLs and IP addressing. **Command:**

```
aws s3api put-public-access-block --bucket <BUCKET_NAME>
```

Replace `<BUCKET_NAME>` with the name of your target bucket.

## Ticket Closure Procedure

To verify fixes and update Jira tickets:

### 1. Verify Fix:

- Confirm that the S3 bucket is encrypted using `aws s3api get-bucket-encryption --bucket <BUCKET_NAME>`
- Verify public access blocks are in place by running `aws s3api put-public-access-block` on a respective bucket.

Replace `<BUCKET_NAME>` with your target bucket name. 2. **Update Jira Ticket:**

Update the status of COMP-101 to RESOLVED and include relevant logs or details, such as:

- Command executed ( `aws s3api put-bucket-encryption` ).
- Logs indicating bucket encryption in progress (e.g., "2023-03-02T17:20:43.523Z – Encryption request to `<BUCKET_NAME>` returned: 200").
- Jira ticket summary and relevant context.

Example Command:

```
aws s3api put-bucket-encryption --bucket <BUCKET NAME> --server-side-encryption-Type AES_256
```

Replace `<BUCKET_NAME>` with the name of your target bucket. 3. **Review and Test (Optional):**

Conduct thorough testing and logging to validate security configurations using AWS CloudWatch logs or other appropriate monitoring tools.

Example:

```
aws s3api get-bucket-encryption --bucket <BUCKET NAME> --verbose
```

Replace ``<BUCKET_NAME>`` with your target bucket name.